



NC STATE
UNIVERSITY

NRP 
Nuclear Reactor Program

NE235 Nuclear Reactor Operations

Module #10

Reactivity Balances & Shutdown Margin

Dept. of Nuclear Engineering
North Carolina State University

Module #10:

Reactivity Balances and Shutdown Margin

Topics:

- A. Absolute Reactivity Balance
- B. Relative Reactivity Balance
- C. Excess Reactivity & Shutdown Margin

Laboratory Session this week:

- Lab#6 - Power Defect Lab

Homework #10 – Due 11/15

Reactivity Balances

Absolute Reactivity Balance

$$\rho_{\text{net}} = \rho_{\text{excess}} + \rho_{\text{rods}} + \rho_{\text{mod temp}} + \rho_{\text{fuel temp}} + \rho_{\text{voids}} + \rho_{\text{xenon}} + \rho_{\text{experiments}} + \rho_{\text{source}}$$

α_{power} = Power Coefficient

Where,

ρ_{excess} = Reactivity in the fuel above that required to go critical (+)

ρ_{rods} = Net worth of all rods in the core (-)

$\rho_{\text{mod temp}}$ = $\alpha_{T_{\text{mod}}} \times T_{\text{mod}} \leftarrow$ Used during startup = **-3.9pcm/°F**

ρ_{fuel} = $\alpha_{T_{\text{fuel}}} \times T_{\text{fuel}} = -303 \frac{\text{pcm}}{\text{MW}}$ or **-1.6 pcm/°F_{fuel}**

ρ_{voids} = $\alpha_{\text{voids}} \times V(\text{cc}) = \textbf{-1.6 pcm/cm}^3$

ρ_{power} = $\alpha_{\text{power}} \times \% \text{Power} \leftarrow$ Used during power ops = **-330pcm/MW**

ρ_{xenon} = Time Dependent based on operating history

ρ_{exper} = Experiments (draining of beamports (-), etc.)

ρ_{source} = **-16pcm** If source is OUT, **0 pcm** if IN

Absolute Reactivity Balance

Example #1 – Absolute Balance:

$$\rho_{\text{net}} = \rho_{\text{excess}} + \rho_{\text{rods}} + \rho_{\text{mod temp}} + \rho_{\text{fuel temp}} + \rho_{\text{voids}} + \rho_{\text{xenon}} + \rho_{\text{exp}} + \rho_{\text{source}}$$

- a) Reactor is loaded with +2100 pcm ρ_{excess} from fuel.
- b) Rods are partially withdrawn leaving –1800 pcm inserted in core.
- c) There is –200 pcm Xenon in core.
- d) Temperature is 90°F (+20°F relative to “Cold Clean” Conditions at 70°F).

∴ What is ρ_{net} at current condition?

$$\rho_{\text{net}} = \rho_{\text{excess}} + \rho_{\text{rods}} + \rho_{\text{xenon}} + \rho_{\text{mod temp}}$$

Where: $\rho_{\text{mod temp}} @ 70^\circ\text{F} = 0 \text{ pcm}$

∴ If $\Delta T_{\text{mod}} = +20^\circ\text{F}$ and $\alpha_{T_{\text{mod}}} = -3.9 \text{ pcm}/^\circ\text{F}$

$$\rho_{\text{mod temp}} = -3.9 \text{ pcm}/^\circ\text{F} \times (+20^\circ\text{F}) = -78 \text{ pcm}$$

$$\therefore \rho_{\text{net}} = +2100 - 1800 - 200 - 78 = +22 \text{ pcm}$$

Absolute Reactivity Balance

Example #1 Continued:

Problems with “Absolute” Reactivity Balance:

Errors/uncertainties in

- ☐ Rod worth curves
- ☐ True core excess
- ☐ Xenon worth curves

These errors and uncertainties make it difficult to accurately calculate ρ_{net} .

$\therefore$ We routinely use the relative reactivity balance method.

Relative Reactivity Balance

Relative Reactivity Balance: (A.K.A. Incremental Change Method)

$$\Delta\rho_{\text{net}} = \Delta\rho_{\text{excess}} + \Delta\rho_{\text{rods}} + \Delta\rho_{\text{mod temp}} + \Delta\rho_{\text{fuel temp}} + \Delta\rho_{\text{voids}} + \Delta\rho_{\text{xenon}} + \Delta\rho_{\text{exp}} + \Delta\rho_{\text{source}}$$

Reference from the last known 'Actual Critical Position' (ACP) where

$$\rho_{\text{net}} = 0$$

Then, find what reactivity parameters have changed since last ACP and make corrections.

Used for calculating 'Estimated Critical Position' (ECP) at startup to project critical rod height, shutdown margin and excess reactivity, and establish that reactivity parameters are sufficiently well known to allow for startup.

If ACP is greater than +/- 200 pcm from ECP, shutdown reactor!

Relative Reactivity Balance

So, if:

$$\Delta\rho_{\text{net}} = \Delta\rho_{\text{excess}} + \Delta\rho_{\text{rods}} + \Delta\rho_{\text{mod temp}} + \Delta\rho_{\text{fuel temp}} + \Delta\rho_{\text{voids}} + \Delta\rho_{\text{xenon}} + \Delta\rho_{\text{exp}} + \Delta\rho_{\text{source}}$$

And, if: $\rho_{\text{net}} = 0$ at both ECP and ACP (i.e. at criticality), then $\Delta\rho_{\text{net}} = 0$

So, Solve for $\Delta\rho_{\text{rods}}$ to obtain ECP where:

$$\Delta\rho_{\text{net}} = 0 = \Delta\rho_{\text{excess}} + \Delta\rho_{\text{rods}} + \Delta\rho_{\text{mod temp}} + \Delta\rho_{\text{fuel temp}} + \Delta\rho_{\text{voids}} + \Delta\rho_{\text{xenon}} + \Delta\rho_{\text{exp}} + \Delta\rho_{\text{source}}$$

$\Delta\rho_{\text{excess}}$ is usually small between startups and we could calculate it based on burnup, but we can assume this term is negligible ($\therefore \Delta\rho_{\text{excess}} = 0$)

$\Delta\rho_{\text{fuel temp}} = 0$ at startup (below the 'Point of Adding Heat')

$$\Delta\rho_{\text{voids}} = 0$$

So:
$$\Delta\rho_{\text{rods}} = -\Delta\rho_{\text{mod temp}} - \Delta\rho_{\text{xenon}} - \Delta\rho_{\text{exp}} - \Delta\rho_{\text{source}} \quad (\text{Eq 2.25})$$

Where: $\Delta\rho = \rho_f - \rho_i$

($\Delta\rho = \text{final reactivity} - \text{initial reactivity}$)

Where @ ECP: $\rho_{\text{rods}f} = \rho_{\text{rods}i} + \Delta\rho_{\text{rods}}$

GIVES ECP

FROM LAST ACP

CALC FROM
ABOVE

Relative Reactivity Balance

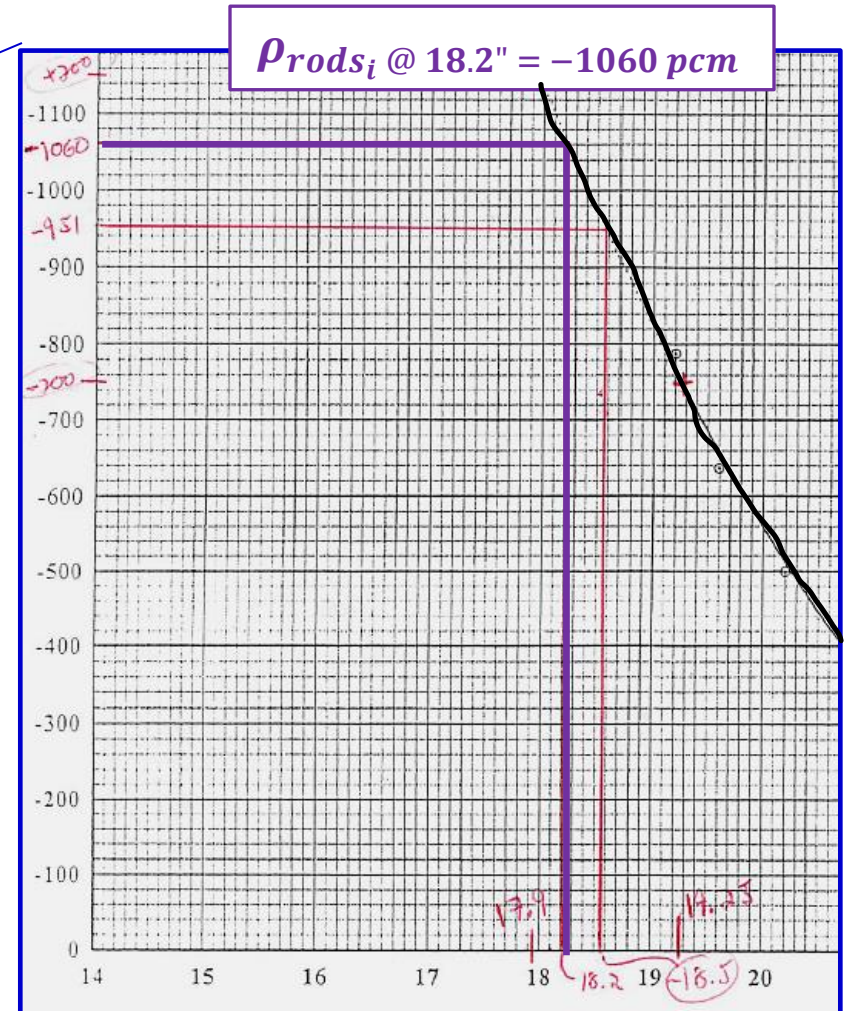
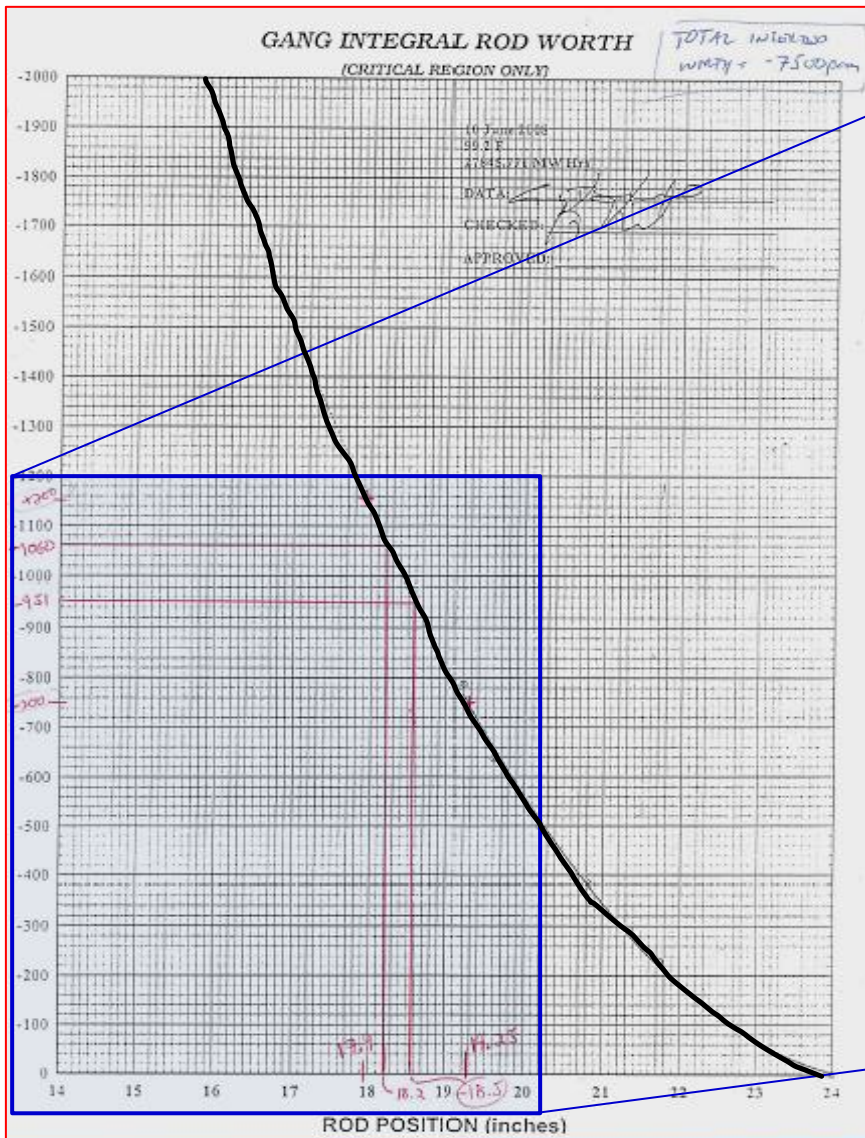
Example 2: Incremental Change Method of Reactivity Balance

Calculate ECP (ρ_{rods_f}) on Tuesday morning @8AM given the following initial conditions at startup on Monday:

- ❑ ACP at Startup on Monday = **18.2" on gang**. From 2008 integral gang rod worth curves (see Slide #9), $\rho_{rods_i} = -1060 pcm$
- ❑ No Xenon at startup $\therefore \rho_{xenon_i} = 0 pcm$
- ❑ Start up source is **IN** at startup $\therefore \rho_{source_i} = 0 pcm$
- ❑ Pool (Mod) Temp at startup $T_{mod_i} = 93^\circ F$
- ❑ Experiments at startup (BT#5 drained): $\rho_{experiments_i} = -35 pcm$

Operations: Then operated for 6 hours on Monday at 500kW and shutdown at 4 pm.

Integral Gang Rod Worth Curve (2008)



Example #2 – continued; Calculate ECP

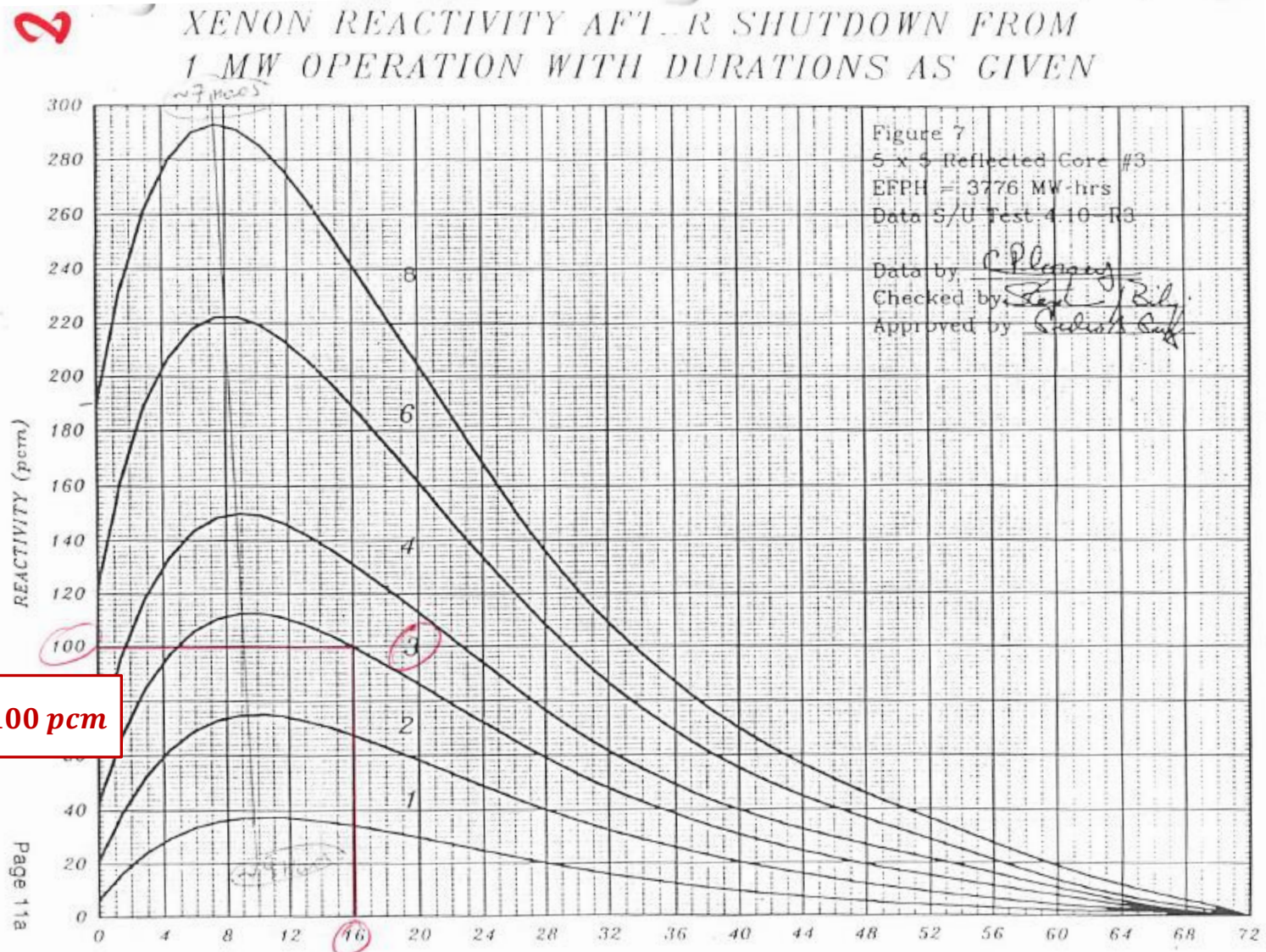
Conditions (**final**) prior to Startup @ 8 am on Tuesday:

Calculate each $\Delta\rho = \rho_f - \rho_i$

- ❑ **Pool Temp:** $T_f = 100^\circ\text{F} \therefore \Delta T_{mod} = T_f - T_i = 100 - 93 = +7^\circ\text{F}$
 $\therefore \Delta\rho_{mod} = \alpha_{T_{mod}} \times \Delta T_{mod} = -3.9 \frac{\text{pcm}}{^\circ\text{F}} \times 7^\circ\text{F} = -27.3 \text{ pcm}$
- ❑ **Xenon:** Operated 6 hours at 500kW = 3 MW-hours and Decayed 16 hrs (see Slide #11 for Xenon Worth Curves) $\therefore \rho_{xeff} = -100 \text{ pcm}$
 $\therefore \Delta\rho_{xenon} = \rho_{xeff} - \rho_{xei} = (-100 - 0) = -100 \text{ pcm}$
- ❑ **Experiments:** Flooded BT#5 $\therefore \rho_{expf} = 0$
 $\therefore \Delta\rho_{exp} = \rho_{expf} - \rho_{expi} = 0 - (-35 \text{ pcm}) = +35 \text{ pcm}$
- ❑ **Startup Source Out** $\therefore \rho_{sourcef} = -16 \text{ pcm}$
 $\therefore \Delta\rho_{source} = \rho_{sourcef} - \rho_{sourcei} = (-16 \text{ pcm} - 0) = -16 \text{ pcm}$

Xenon Worth Curves

ρ_{xenon} (pcm) as function of MW-hrs and decay since shutdown



Example #2 - continued; Calculate ECP

Solve for $\Delta\rho_{\text{rods}}$ where $\Delta\rho_{\text{rods}} = -\Delta\rho_{\text{mod temp}} - \Delta\rho_{\text{xenon}} - \Delta\rho_{\text{exp}} - \Delta\rho_{\text{source}}$

$$\text{So, } \Delta\rho_{\text{rods}} = \underbrace{-(-27.3)}_{\Delta\rho_{\text{mod temp}}} - \underbrace{(-100)}_{\Delta\rho_{\text{xenon}}} - \underbrace{(+35)}_{\Delta\rho_{\text{exp}}} - \underbrace{(-16)}_{\Delta\rho_{\text{source}}} = 108.3 \text{ pcm} = \Delta\rho_{\text{rods}}$$

Now, solve for $\rho_{\text{rods final}}$:

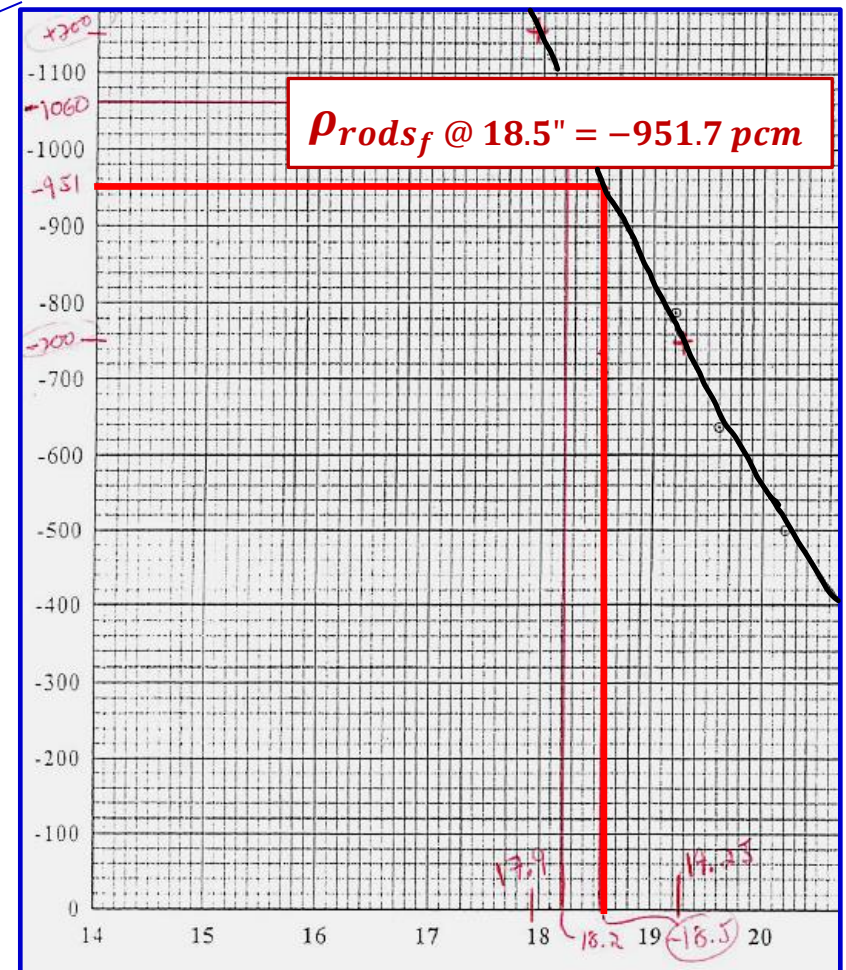
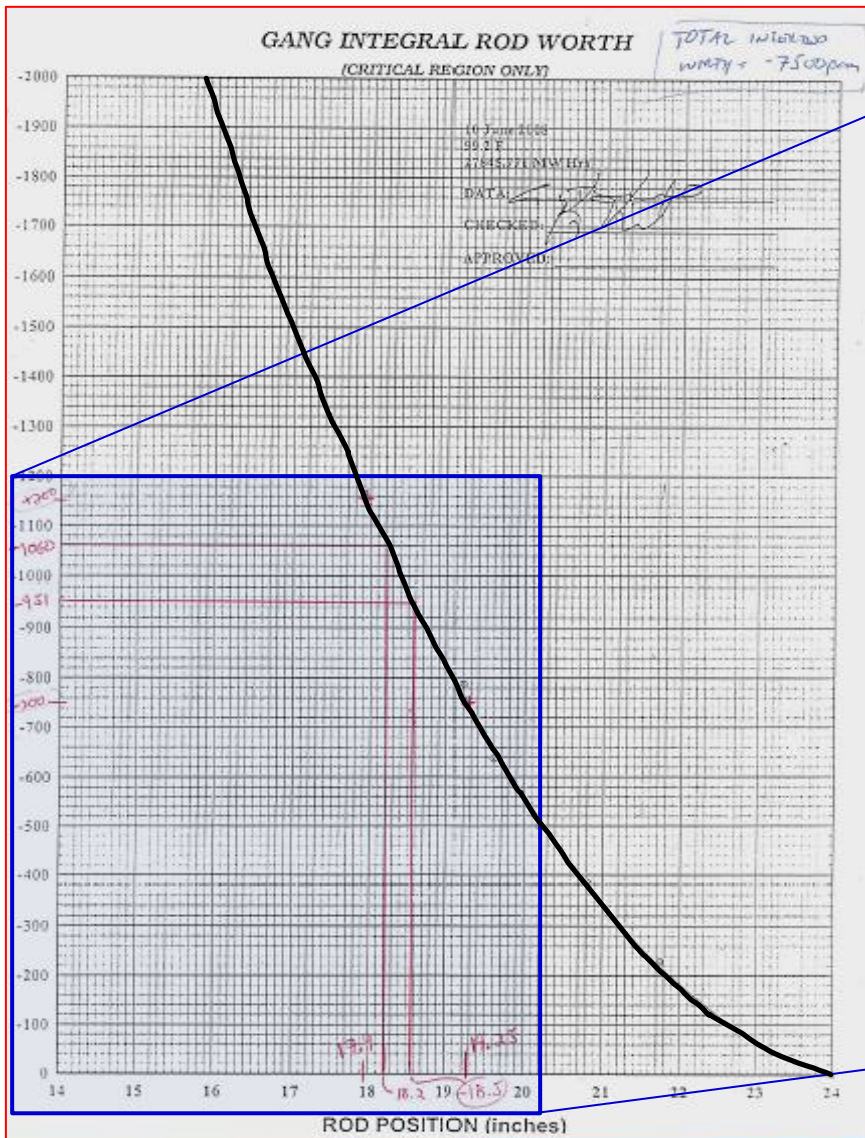
Where: $\Delta\rho = \rho_f - \rho_i \quad \therefore \rho_{\text{rods}_f} = \rho_{\text{rods}_i} + \Delta\rho_{\text{rods}}$

So: $\rho_{\text{rods}_f} = -1060 \text{ pcm} + 108.3 \text{ pcm} = -952 \text{ pcm}$

$\therefore \text{ECP} = 18.5''$ on gang from curves (see Slide #13)

+/- 200 pcm = 17.9'' to 19.25''

Integral Gang Rod Worth Curve (2008)



ECP +/- 200 pcm band = 17.9" – 19.25"

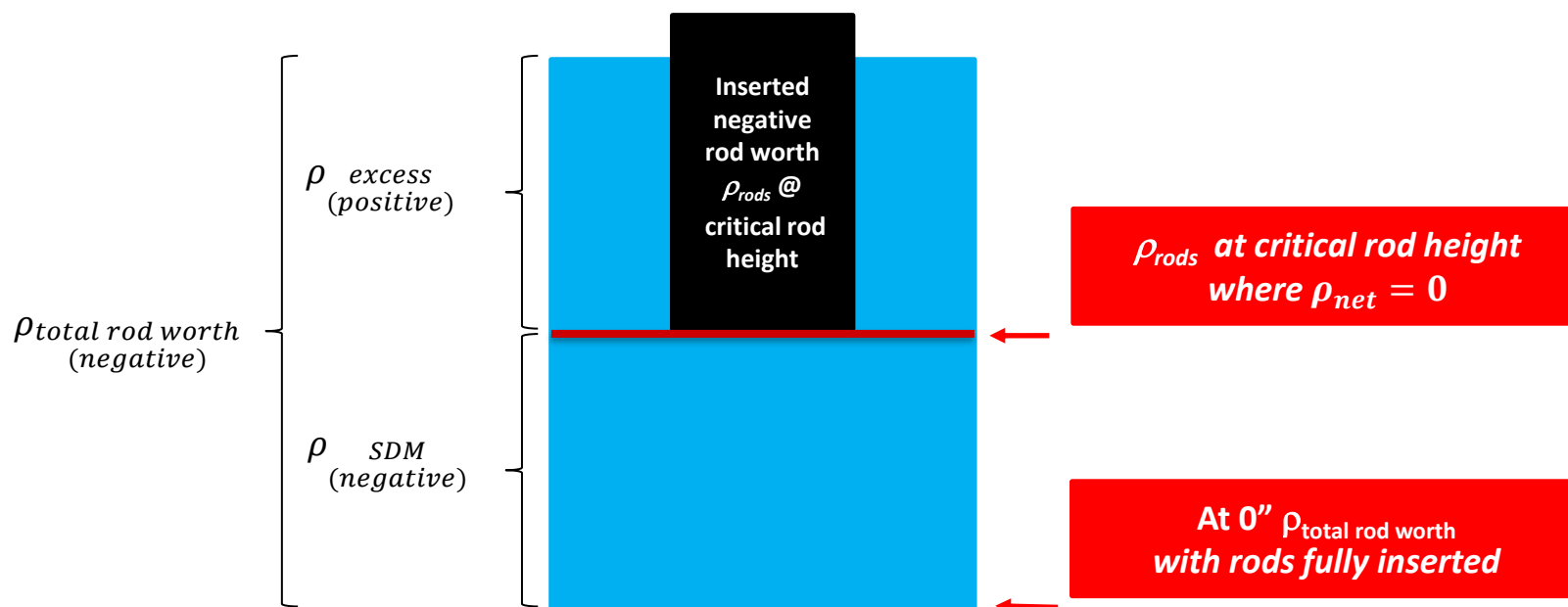
Excess Reactivity & Shutdown Margin

ρ_{excess} = positive reactivity in core above that required to go critical

$\rho_{\text{Shutdown Margin (SDM)}}$ = negative reactivity available to shutdown the reactor

Reference both to cold ($T_{\text{mod}} = 70^\circ\text{F}$), clean ($\rho_{\text{Xe}} = 0$), critical conditions.

At cold, clean, critical conditions: $\rho_{\text{excess}} = -(\rho_{\text{rods}})$



Where: $\rho_{\text{SDM}} = \rho_{\text{total rod worth}} + \rho_{\text{excess}}$

Example #3 – Excess Reactivity & Shutdown Margin

Example 3: Now, calculate excess reactivity and shutdown margin on Tuesday morning @8AM corrected to cold, clean critical conditions given the following reactivity conditions:

ACP Reactivity conditions Tuesday morning @8AM (*redefine initial conditions @ ACP*):

$$\text{Pool (Mod) Temp} = 100^{\circ}\text{F} \ (T_{mod\ i})$$

$$\rho_{xenon} = -100\text{ pcm} \ (\rho_{xenon\ i})$$

$$\rho_{experiments} = 0\text{ pcm} \ (\rho_{exp\ i})$$

$$\text{Start up source is out} \therefore \rho_{source} = -16\text{ pcm} \ (\rho_{source\ i})$$

$$\rho_{rods} = -952\text{ pcm} \ (\rho_{rods\ i})$$

Definition of Cold-Clean-Critical Conditions:

$$\text{Pool (Mod) Temp} = 70^{\circ}\text{F} \ (T_{mod\ f})$$

$$\text{No Xenon} \therefore \rho_{xenon} = 0\text{ pcm} \ (\rho_{xenon\ f})$$

$$\text{No Experiments} : \rho_{experiments} = 0\text{ pcm} \ (\rho_{exp\ f})$$

$$\text{Start up source is in} \therefore \rho_{source} = 0\text{ pcm} \ (\rho_{source\ f})$$

Example #3 – Excess Reactivity & Shutdown Margin

So, calculate each $\Delta\rho = \rho_f - \rho_i$ correcting to cold clean critical conditions on Tuesday morning:

- 1. Pool Temp:** $\Delta T_{mod} = T_f - T_i = 70 - 100 = -30^\circ\text{F}$
 $\therefore \Delta\rho_{mod} = \alpha_{T_{mod}} \times \Delta T_{mod} = -3.9 \frac{\text{pcm}}{^\circ\text{F}} \times -30^\circ\text{F} = +117 \text{ pcm}$
- 2. Xenon:** $\rho_{Xef} = 0 \text{ pcm},$
 $\therefore \Delta\rho_{xenon} = \rho_{Xef} - \rho_{Xei} = 0 - (-100 \text{ pcm}) = +100 \text{ pcm}$
- 3. Experiments:**
 $\therefore \Delta\rho_{exp} = \rho_{expf} - \rho_{expi} = (0 - 0) = 0 \text{ pcm}$
- 4. Startup Source Out:**
 $\therefore \Delta\rho_{source} = \rho_{sourcef} - \rho_{sourcei} = 0 - (-16 \text{ pcm}) = +16 \text{ pcm}$

Example #3 – Excess Reactivity & Shutdown Margin

Solve for $\Delta\rho_{\text{rods}}$ where $\Delta\rho_{\text{rods}} = -\Delta\rho_{\text{mod temp}} - \Delta\rho_{\text{xenon}} - \Delta\rho_{\text{exp}} - \Delta\rho_{\text{source}}$

$$\text{So, } \Delta\rho_{\text{rods}} = -(\underbrace{+117}_{\Delta\rho_{\text{mod temp}}}) - (\underbrace{+100}_{\Delta\rho_{\text{xenon}}}) - (\underbrace{0}_{\Delta\rho_{\text{exp}}}) - (\underbrace{+16}_{\Delta\rho_{\text{source}}}) = -233 \text{ pcm} = \Delta\rho_{\text{rods}}$$

Now, solve for $\rho_{\text{rods final}}$:

Where: $\Delta\rho = \rho_f - \rho_i \therefore \rho_{\text{rods } f} = \rho_{\text{rods } i} + \Delta\rho_{\text{rods}}$

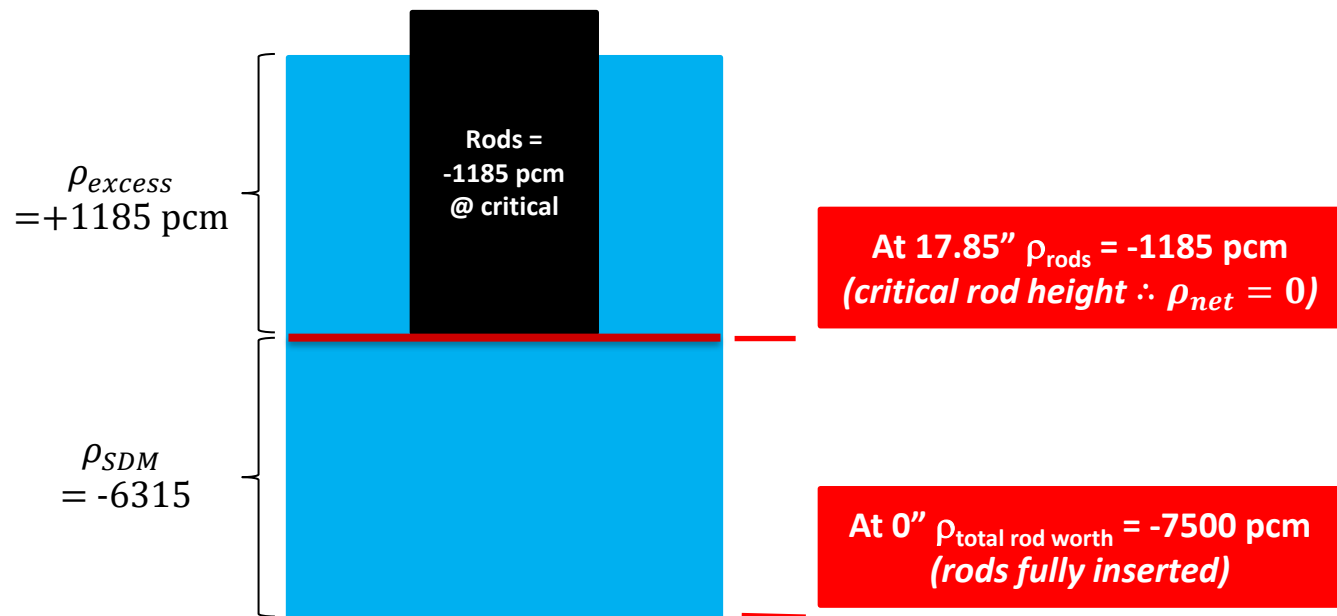
So: $\rho_{\text{rods } f} = -952 \text{ pcm} + (-233) \text{ pcm} = -1185 \text{ pcm}$

$\therefore \text{ECP} = 17.85''$ on gang from curves (see Slide #13)
 at cold, clean critical conditions

Example #3 – Excess Reactivity & Shutdown Margin

Where at cold, clean, critical: $\rho_{excess} = -(\rho_{rods})$

So: $\rho_{excess} = -(\rho_{rods}) = -(-1185) = +1185\text{pcm}$



Where: $\rho_{SDM} = \rho_{total\ rod\ worth} + \rho_{excess}$

Shutdown Margin (ρ_{SDM}) = $(-7500) + (+1185) = -6315\text{pcm}$

Reactivity Balances

Importance of Reactivity Balances:

1. Know ECP within ± 200 pcm of ACP prior to startup – *otherwise you don't know what is going on inside of reactor, so **SHUTDOWN!***
2. Know your estimated critical rod position (ECP) at cold-clean-critical to determine compliance with Technical Specifications “Limiting Condition for Operations” (LCO) Requirements PRIOR TO STARTUP:
 - a) TS §3.2(b) Excess Reactivity $< 3.97\% \Delta K/K$ (3970 pcm)
 - b) TS §3.2(a) Shutdown Margin $> 0.4\% \Delta K/K$ (400 pcm)with:
 - i. Highest Scrammable Rod Fully Withdrawn
 - ii. Shim Rod Fully Withdrawn
 - iii. Experiments at Their Most Reactive Condition
 - iv. Relative to “Cold, Clean, Critical” (*Xenon free; $T_{mod} = 70^\circ\text{F}$*)

Reactivity Balances

Why have these limits?

- a) **Shutdown Margin:** To assure that the reactor may always be shutdown from any condition of operation and will remain shutdown after cooldown, xenon decay, or removal of an experiment.
- b) **Excess Reactivity:** An upper limit on excess reactivity is additional assurance that an adequate shutdown margin is maintained. It also limits the extent of a power excursion in the event of a transient accident.